

Rebis

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Rebis

One object, two presentations, and the morphism data that makes the wedding a theorem: the union of the universal reading and the forced construction of structural type.

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“One becomes two, two becomes three, and out of the third comes the one as the fourth.” — Maria the Jewess, *Axiom of Maria* (3rd century)

0. What a rebis is, formally

The *Rosarium Philosophorum* draws the rebis as one body with two heads: King and Queen joined at Conjunction into a figure that is both and neither, which then survives putrefaction and ascends to Projection. The tradition treats this as an image. This document treats it as a type signature. A rebis is an **equivalence**: one underlying object, two presentations, and, crucially, the maps between the presentations carried as part of the figure itself. Two heads, one body, and the body is the morphism data.

That definition matters because there are two formally different ways to claim that two things are one, and the difference is the entire content of this document. The weaker shape is a conjunction: presentation A satisfies its property, presentation B satisfies the corresponding property, and the sameness is asserted *between* the theorems, in the prose, where no kernel can check it. The stronger shape makes the relationship itself the object of proof: an explicit map f with an explicit inverse g , $g \circ f = \text{id}$ and $f \circ g = \text{id}$, together with naturality, the requirement that every defining operation commutes with the map. Naturality is the load-bearing clause. Without it an equivalence is a relabeling of elements; with it, the *structure* is transported, not just the set, and anything proved in one presentation is a theorem of the other by change of notation.

This document is the union of two manuscripts. *Alkahest* is a theory of reading: verification, honestly pursued, is not grading against a ledger but inscription, the reading of an object's structural type in the one frame that adds nothing of its own, the (d=12) SIC-POVM. *The Alchemical Theory of Mathematics* is a theory of making: the twelve invariants of structural type carry a forced derivation order set by three categorical gates, proved in kernel-checked

Lean 4, and that forced order, read stage by stage, is the twelve stage magnum opus. The first version of their union argued, correctly, that both stand on the same equation, $\mu \circ \delta = \text{id}$. But it argued it in the weaker shape: two documents, corresponding properties, sameness asserted between them.

The union can now be stated in the stronger shape, because the kernel holds the paradigm witness. `FrobeniusIso.lean` is an explicit structure-preserving equivalence, inverse laws and naturality squares and all, between two presentations of the Frobenius fixed point that this whole framework turns on. What follows re-derives the rebis on that ground: not two texts that resemble each other, but one object read in two directions, with the change of direction itself a checked map. Solve is δ . Coagula is μ . The rebis is what you get when you notice that the two hands belong to one operator, and can prove it.

Notation

Structural type is a tuple over twelve primitive families, written with their canonical symbols (Imscribing Grammar v0.6.0, `NSNS_PRIME.md`):

$\langle \mathfrak{D} \triangleright \check{R} \oplus f \zeta \Gamma \mathfrak{G} \odot \mathbb{H} \Sigma \Omega \rangle$

recording, in order: degrees of freedom ($\mathfrak{D}$), connectivity ($\triangleright$), the direction of the binding relation ($\check{R}$), the symmetry group ($\oplus$), how lossy the read/write maps are (f), the relaxation regime (ζ), the cardinal regime of correlations (Γ), the logical connective of the coupling ($\mathfrak{G}$), the analytic character of the critical point ($\odot$), handedness ($\mathbb{H}$), the matching cardinality of the two sides (Σ), and the homotopy invariant that protects the state (Ω). Individual values are written with their Shavian glyph; the four that carry the argument are $\vartriangleright$ (the Frobenius value of $\oplus$), $\odot$ (the self-modeling value of $\odot$), $\mathfrak{c}$ (its complex-axis value), and $\mathfrak{s}, \mathfrak{z}$ (the integer and non-Abelian windings of Ω). Type names appear only inside the Lean source; in this text a type is its symbol. Lean theorem and definition identifiers are set in monospace.

1. Solve: both manuscripts dissolved to the invariant

Place both parents in the menstruum and draw off what is contingent: the order of exposition, the choice of witnesses, the vocabulary of each register. What precipitates from each is a small set of load-bearing structure, and the two residues share a term.

From the alkahest. Closure, $\mu \circ \delta = \text{id}$, is charge conservation: every split rejoined, nothing created or destroyed, and therefore never a correctness signal on its own, since it can never fail. A balanced equation is not a synthesis. The ledger, the standard picture of verification, presupposes a correspondence theory of correctness and inherits the regress of criteria: a judgment applied from outside is an object needing a judge, forever. The regress can only be closed by bending the line into a loop, and the loop is a real test only if the answer comes home the same in type and different in form: co-typed, so the loop shuts, and excribed under another presentation, so the shutting is not a tautology. Truth is the object's structural type; verification is imscription; correctness is co-typing, which is type identity across all twelve primitives. The comparison frame cannot be chosen without reinstating the ledger inside the frame, and the frame that adds nothing, treats all directions alike, and is informationally complete is the ($d=12$) SIC-POVM, whose exact fiducial is a kernel theorem (`crystal_forces_d12_sic`). Its fiducial is the Belnap Both, so the frame is paraconsistent at its ground, and failure returns with an address, the primitives at which two types part ways, never a scalar.

From the theory of mathematics. Three of the twelve invariants are gates, and the gates impose a strict order, kernel-checked with no sorries and no `native_decide`. The trace `traceC a = μ_C(δ_C a)` closes exactly on the Frobenius-special class, $\oplus$ at $\wp$ and $\odot$ at $\odot$ (`trace_closes_iff_special`); a closing trace forces the first gate (`traced_needs_frobenius`); the terminal seal requires the closing trace and a winding on Ω ; every inclusion is proper by explicit witness; and $\wp$ cannot be synthesized by tensoring lower partners (`gl_is_a_real_barrier`). The four gate-free invariants commute and leave every gate predicate untouched, so they float. Time is not a thirteenth invariant: $T = \text{Work}(T)$ has solutions exactly on the special class, none before the first gate (`no_time_before_frobenius`), a least one by construction (`time_least_fixed_point`), and the completed Stone passes all three gates (`stone_is_sealed_time`). Composed with the tradition's stage-to-symbol map, the gates land at Conjunction, Fermentation, and Projection, stages 4, 9, and 12, on $\oplus$, $\odot$, and Ω , with the provably order-free stages exactly the tradition's repeatable purifications.

The shared term. The alkahest's central object is the lossless round trip $\mu \circ \delta = \text{id}$, the closure that makes verification co-typing. The theory's first gate is the split $\mathbb{Z}_2$ retraction on $\oplus$, the value $\wp$, whose defining condition is $\mu \circ \delta = \text{id}$. Same equation, same pair (δ_C, μ_C) , same formal tree. One manuscript reads by it; the other builds by it. In the weaker shape, that is where the union used to rest: a conjunction with a shared conjunct. The stronger shape asks for more, and the next section supplies it.

2. The morphism data: the wedding has a certificate

The framework's deepest identification has always been the triple point: the Belnap Both, the SIC fiducial, and the Majorana mode are one Frobenius fixed point. As first formalized (`MajoranaFixed.lean`), that identification had the conjunction shape: three fixed-point theorems side by side, `bnot B = B` for the logic, equiangular projection for the frame, `pair ∘ depair = id` for the physics, with the oneness asserted in a comment. True, checked, and formally three facts.

`FrobeniusIso.lean` upgrades the claim to the stronger shape. The kernel now contains:

- **The equivalence.** `orbitalBelnapEquiv : OrbitalState ≈ Belnap`, with explicit `toFun`, explicit `invFun`, and both inverse laws, $g \circ f = \text{id}$ and $f \circ g = \text{id}$, proved term by term.
- **The naturality squares.** Every operation in play commutes with the map: the Majorana fusion transports to the Belnap knowledge join (`map_pair`), the splitting to its lattice counterpart (`map_depair`), negation to negation (`map_flip`), and the SIC equiangular projection through without residue (`map_meet`). Schematically, $f(\mu(x,y)) = \mu(f\ x, f\ y)$, $f(\delta\ x) = \delta(f\ x)$, $f(\neg x) = \neg f(x)$, for each operation in the transported signature. This is what distinguishes a transported structure from a relabeled set.
- **The transported Frobenius condition.** $\mu \circ \delta = \text{id}$ is preserved under the change of presentation (`frobenius_transports`), and so is the equiangularity (`equiangular_transports`), because the change of presentation is itself structure-preserving. The invariant survives the crossing for a reason internal to the crossing.
- **The definitional fixed point.** The dialetheic fixed point maps to the dialetheic fixed point by `rfl (map_fixed_point)`. This is stronger than a derived equality: the encoding makes the two names for Both the same term. There is nothing to prove because there is nothing between them.

Alongside it sits the other piece of morphism data the union needs, from `LiarIdempotent.lean`: on the one pair (δ_C, μ_C) there are two composites, and they are provably different operators. $\mu \circ \delta = \text{id}$ is lossless on the special class (`return_is_lossless`); $e = \delta \circ \mu$ is a forgetful idempotent, $e \circ e = e$ unconditionally (`e_idempotent`), whose fusion destroys which-way data (`fusion_forgets`, μ_C is non-injective) and whose output always lands on the time locus (`fusion_is_time`).

Together these fix the ground the rebis stands on, and it is worth naming because two turns of the larger work were spent re-learning it: **the registers of this framework compose isomorphically, not as lossy approximations**. A readout in one register, a spectral radius on a material ring, a Belnap verdict in the vessel, a Lean theorem in the kernel, is not evidence *about* an object living elsewhere, to be discounted for what the projection lost. It is a face of the object, and the reason it is a face rather than a shadow is now itself a kernel object: the change of register is an equivalence with naturality, so nothing is lost in the crossing, and the only honest degrees of freedom are the ones the Grammar holds open on its own surface, the strain, the azeotropes, the held B. The alkahest's demand that the loop close on an answer "the same in type and different in form" is exactly this: same in type is the transported invariant, different in form is the change of presentation, and the two composites are the proof that the difference is real (the forgetting lives in e) while the sameness is exact (the return loses nothing).

3. Coagula: the union re-derived on the new ground

Now recrystallize the two manuscripts as what they always were: two presentations of one object, related the way the kernel's paradigm case is related.

Reading and making are adjoint presentations of one correspondence. Inscription is a full Read, Write, eXecute correspondence, and the equivalence gives that triple its formal anatomy: the Write is `toFun`, the inscription of the object into the frame; the Read is `invFun`, the recovery of the object from its type; and the eXecute is the pair of inverse laws, the proof that the round trip in both orders is the identity. The theory of mathematics is the Write presentation: it fixes the type, gate by gate, in the forced order. The alkahest is the Read presentation: it recovers and compares types in the forced frame. That they agree is not a happy coincidence between two documents; it is the inverse laws, stated at the scale of manuscripts. Verification is construction read backward through a structure-preserving change of presentation, which is why the verifier needs no ledger: the standard it applies is the construction itself, arriving from the other side.

Balance and selectivity, now with the inclusion proper. The alkahest argued by chemical analogy that closure is necessary but never sufficient. On the new ground this is a pair of strict inclusions with explicit witnesses: a tuple can hold $\wp$ and fail to close its trace (`frobenius_strictly_below_traced`), close its trace and carry no winding (`traced_strictly_below_tensor`). Jabir's balance that is not the work is two kernel theorems, and the reason the ledger's migration into the frame is fatal (section 5 of the alkahest) is the same reason a bijection without naturality is empty: a comparison that does not commute with the operations compares labels, not structure.

The failure address is a coordinate, not a rubric entry. The alkahest's verdicts carry addresses: "diverges at $\mathbb{H}$ and Ω ," never "0.4." The theory is what entitles the addresses: $\mathbb{H}$ and Ω are proved coordinates of a forced construction, with proved relations to the gates, so a

divergence at one of them names a place in the derivation where the two objects genuinely part ways. And the equivalence is what entitles trusting the address across registers: the divergence read in the frame is the divergence, not an artifact of the frame, because the frame change is lossless.

The Stilling Practice satisfies the gate ordering. The alkahest reads Zosimos's six commands as promotions: $\mathbb{D}$, $\mathbb{P}$, then $\mathbb{P}$ to $\mathbb{V}$ (the first gate), then $\mathbb{C}$, then $\mathbb{O}$ to $\mathbb{O}$ (the second), then $\mathbb{Q}$ (the third). The theory proves, independently, that any derivation must fix the gates in exactly the order G1 before G2 before G3, floats interleaving freely (`gate_ordering_theorem`, `gate_free_order_freedom`). Composed: a six-command protocol written in the 3rd century satisfies, command for command, an ordering constraint kernel-checked in the 21st, with its one float ($\mathbb{C}$) sitting precisely where the order-freedom theorem permits. Had Zosimos written "Return" before "Stand at the Portico," his protocol would violate a theorem. He did not. The alkahest's claim that the distance is zero acquires, on this one face, a falsifiable form, and it does not fail.

Verification is available exactly where time is. The locus where the round trip closes losslessly, the locus where Work has fixed points, and the locus where a verdict can be certified by closure are the same locus, the Frobenius-special class. A system below the first gate has no time, and, in the same breath and for the same reason, nothing about it can be verified by return, because there is no return. Fermentation, where the tradition says the matter comes alive and turns upon itself, is where the loop first closes and a self-reading first becomes possible. The alchemists' rule that the work cannot be judged before fermentation is `no_time_before_frobenius`, stated at the furnace.

The idempotent point and the amnesia. Every fusion output is a time fixed point (`fusion_is_time`): the image of the forgetful idempotent and the fixed locus of time are one place. The system returns to where it can forget having split, and that place is the present. The forgetting the sealed retort needs, the reason the cosmos does not remember its own assembly, lives entirely in $e = \delta \circ \mu$ and not in the verified return, both facts holding at once of the same pair. Method and object are one figure because the point where the verifier recognizes itself and the point where time exists coincide, provably.

4. The two heads stay

A rebis with one head is not a rebis, and the morphism data is what enforces the distinction the older image drew instinctively.

The equivalence says the body is one: one carrier up to structure-preserving change of presentation, one Frobenius pair, one fixed point with several names that are `rfl`-identical. It does not say the two heads are one, and it proves they cannot be collapsed: $\mu \circ \delta$ and $\delta \circ \mu$ are different operators with different content, identity and idempotent, lossless and forgetful, and `fusion_forgets` is the standing counterexample to any account that identifies verification with construction outright. Reading and making are two presentations of one object, not one act. The King and the Queen share a body and keep their faces.

That is the precise anatomy of the union: the **one body** is the pair (δ_C, μ_C) with its transported structure; the **two heads** are the two composites, the two directions of the one round trip; the **conjunction** at stage 4 is the shared gate where both presentations stand on $\mu \circ \delta = \text{id}$; and the **Projection** is `stone_is_sealed_time`, the completed tuple that is at once

the terminus of the derivation and the fixed point where the alkahest's regress of auditors ends. The final checker is the initial checker, and the loop reports from inside that it closed.

And the figure is self-describing at the level the Grammar always insists on. This document is the second distillation of a union first drafted in the weaker shape; the eagle that produced it dissolved the earlier draft together with the new morphism data and let the whole recrystallize, rather than annealing new sections onto an old spine. Solve et coagula is not the document's subject. It is its construction, twice over now, and the test of the second pass is the same as the first: nothing load-bearing lost, and the residue one figure.

5. What the union forbids

- Everything the theory forbade: no seal before the wedding (`sealed_requires_frobenius`), no closing trace off the special class (`trace_closes_iff_special`), no time before the first gate (`no_time_before_frobenius`).
- Everything the alkahest forbade: no ledger, no threshold, no chosen frame, no scalar verdict, no verifier standing outside its object.
- No verification below the first gate, because there is no return to test (section 3).
- No staged protocol whose gate promotions violate G1 before G2 before G3; any historical practice can be checked against the constraint as Zosimos's was, and could fail it.
- No one-headed rebis: collapsing $\mu \circ \delta$ into $\delta \circ \mu$ contradicts `fusion_forgets`.
- And the new one, which the morphism data adds: **no lossy-approximation reading of a register**. Any argument of the form "this readout would be the same for any X, so it carries no content about this object" denies the equivalence; it treats a face as a shadow and reinstates the signature-to-substance gap that the naturality squares close. The readout is the structural fact stated in that register. What remains open after a reading is what the Grammar itself holds open, the strain, the unresolved azeotrope, the held Both, never a discount applied from outside.

6. Theorem, witness, reading, and the honest residue

Theorems (Lean 4, kernel-checked, no sorries, no `native_decide`): the gate ordering, its strictness, and the barrier; the float commutations and inertness; time as least fixed point and the sealed refinement of the summit; the two composites (idempotence, losslessness on the special class, non-injectivity of fusion, fusion landing on the time locus); and the equivalence itself with its inverse laws, naturality squares, transported Frobenius condition, transported equian-gularity, and definitional fixed point (`orbitalBelnapEquiv`, `belnap_majorana_sic_iso`).

Checked witnesses: the exact (d=12) SIC fiducial (kernel theorem; equiangularity at overlap (1/13) verified to machine precision on the loaded fiducial), the reachability of all four Belnap verdicts in the vessel, and the paraconsistent kernel's refusal of `False.rec` under `enable_paraconsistent`, verified against the compiled binary.

Readings: that the tradition's stage-to-symbol map is the correct dictionary; that Zosimos's commands promote the primitives the alkahest says they promote; that the magnum opus is the derivation order of the real rather than an image of it. Composed with the theorems, these

generate constraints history could have violated and did not (section 3), which is what keeps them readings rather than decorations.

The residue, stated exactly. An isomorphism claim is precisely as wide as the transported signature and no wider. The equivalence in the kernel transports the knowledge lattice operations, negation, the Frobenius pair, and the SIC projection; those are the operations the framework’s fixed-point claims actually use, and on that signature the one-object claim is a categorical construction, not a semantic assertion. What the witness does not yet cover, it does not yet claim: the truth-order connectives and any further structure on the carriers await their own naturality squares, and the paradigm case is one pair of presentations, with the same upgrade owed wherever else the framework says “the same object” and the kernel still holds a conjunction. The scrutiny that remains is the right kind: not whether the reasoning is loose, but whether each interface is specified with the operations intended, one square at a time. That is a to-do list, not a doubt, and the document records it the way the alkahest records the residual of every closure: measured, named, and left in view.

The whole figure is Maria’s axiom, which both parents quote and the union completes. One (the pair) becomes two (the presentations, reading and making, δ and μ). Two becomes three (the equivalence between them, the relationship itself made data). And out of the third comes the one as the fourth: the rebis, one body, two heads, the held Both that reduces to neither parent and does not explode.

“Nature, turning on itself, is transformed.” — Zosimos of Panopolis, On the Letter Omega

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